EEE523 – Inertial Navigation Systems

Project II

Implementation of GNSS / 3D RISS Integration using Loosely Coupled Integration

Background:

GNSS and INS have complementary characteristics. INS suffers from various systematic as well as stochastic errors which grow with time. Consequently, navigation solution drifts with passage of time and eventually becomes useless. In contrast, GNSS errors have noise-like behavior and do not grow with time. Therefore, to limit the INS errors, INS is integrated with GNSS in variety of methods. These integration methods are loosely coupled, tightly coupled and ultra-tightly coupled. You will employ the loosely coupled INS/GNSS integration scheme to fuse the INS and GNSS solutions to obtain a navigation solution which is better than either INS or GNSS can provide when operating as stand-alone.

Data provided.

You will be provided with the trajectory data for the following IMUs:

1. High-end tactical (HG1700, Honeywell)
2. Low cost (automotive grade) MEMS-based (IMU CC-300, Crossbow)

The data will contain the following:

1. Raw IMU data containing the specific force measurements from accelerometers and rotation rates from gyroscope.
2. Vehicle velocity data obtained through odometer.
3. GNSS position and velocity data
4. Reference data from High-end tactical which will serve as truth data for position, velocity and attitude

Requirements:

This project involves the implementation of GNSS and reduced inertial sensor system (RISS) integration using loosely-coupled integration algorithm. The student is required to carry out the following:

1. Familiarize yourself with the raw GNSS, INS data and odometer data.
2. Familiarize yourself with the Loosely coupled GNSS/RISS integration algorithms
3. Implement the GNSS/3D RISS integration using Loosely coupled (LC) GNSS/INS integration.

4. Plot the output of the algorithm (position, velocity and azimuth), the reference solution and the error between them for the whole trajectory (set of data points).
5. Incorporate a procedure to introduce GNSS signal blockage and analyze the output of the algorithm during simulated GNSS outage.
6. Write a report that should include following:
    a. Brief overview full (six inertial sensors) GNSS/INS integration and their pros and cons.
    b. Comment on advantages of GNSS/RISS integration and its limitations.
    c. Discuss the results for various parts of the project including (but not limited to):
        i. Accuracy of the algorithm as compared to the reference.
        ii. Effects of introducing the outages at different phases of the trajectory.
        iii. Effects of change of KF parameters on the results
    d. Suggest possible improvements to enhance the performance of the algorithm

References:

Check Paper5 in the useful papers folder.